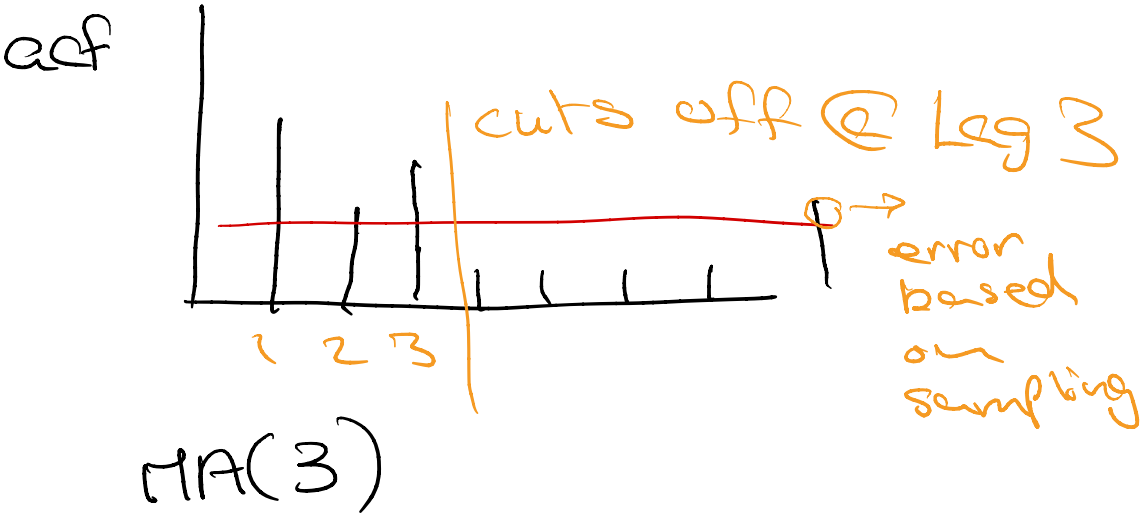
$$\varepsilon_t = y_t - f_t$$

error at time t

actual at time t

forecasted value for ts at time t which we do at time k

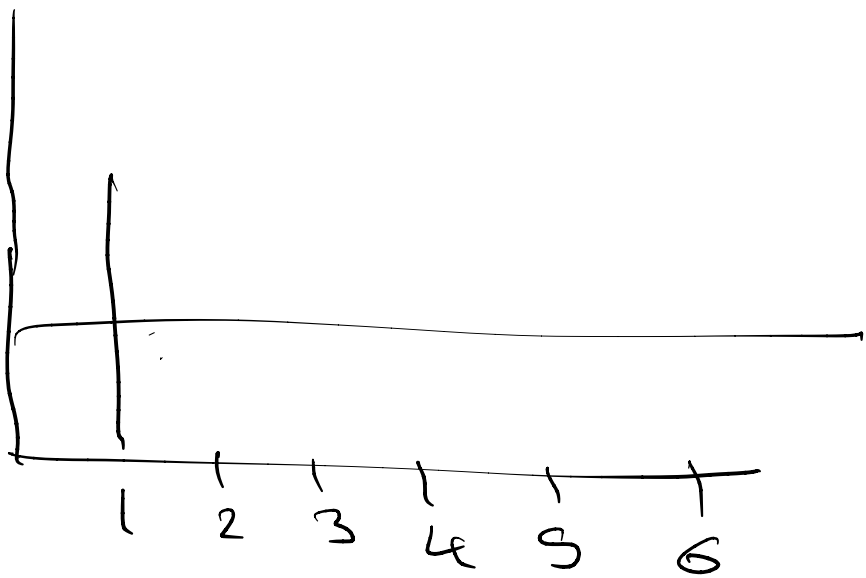
$k < t$



acf

die down
or exponential
decay

pacf



AR(1)

AR(1) model :

$$y_t = \delta + \phi y_{t-1} + \varepsilon_t$$

y_t → value of my ts at time t

δ → Level value

ϕ → regression coefficient which essentially captures the autocorrelation at Lag 1

y_{t-1} → previous value of ts

ε_t → error term

The model above looks a whole lot like :

$$y_t = \delta + y_{t-1} + \varepsilon_t$$

↳ random walk model with a drift

$$\underbrace{\rho(k)}_{\text{auto correlation fn}} = \phi_1 \underbrace{\rho(k-1)}_{\text{autocorrelation at Lag 1}} + \phi_2 \rho(k-2)$$

ϕ_1: regression weight for y_{t-1}

MA(q):

$$y_t = \mu + \varepsilon_t + \boxed{}$$

μ → Level of TS

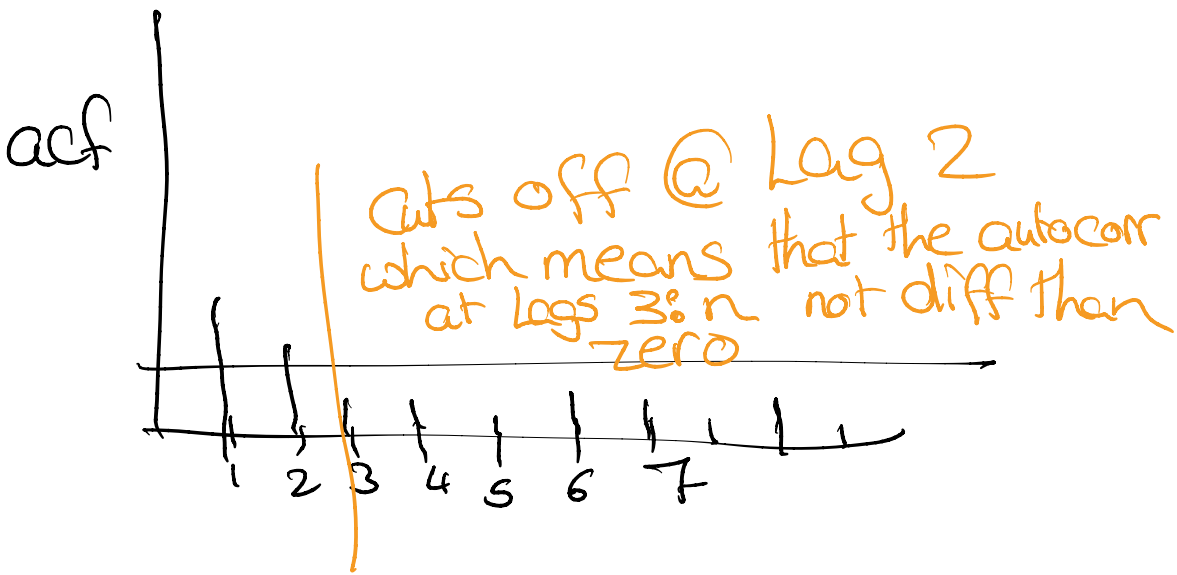
ε_t → error at time t

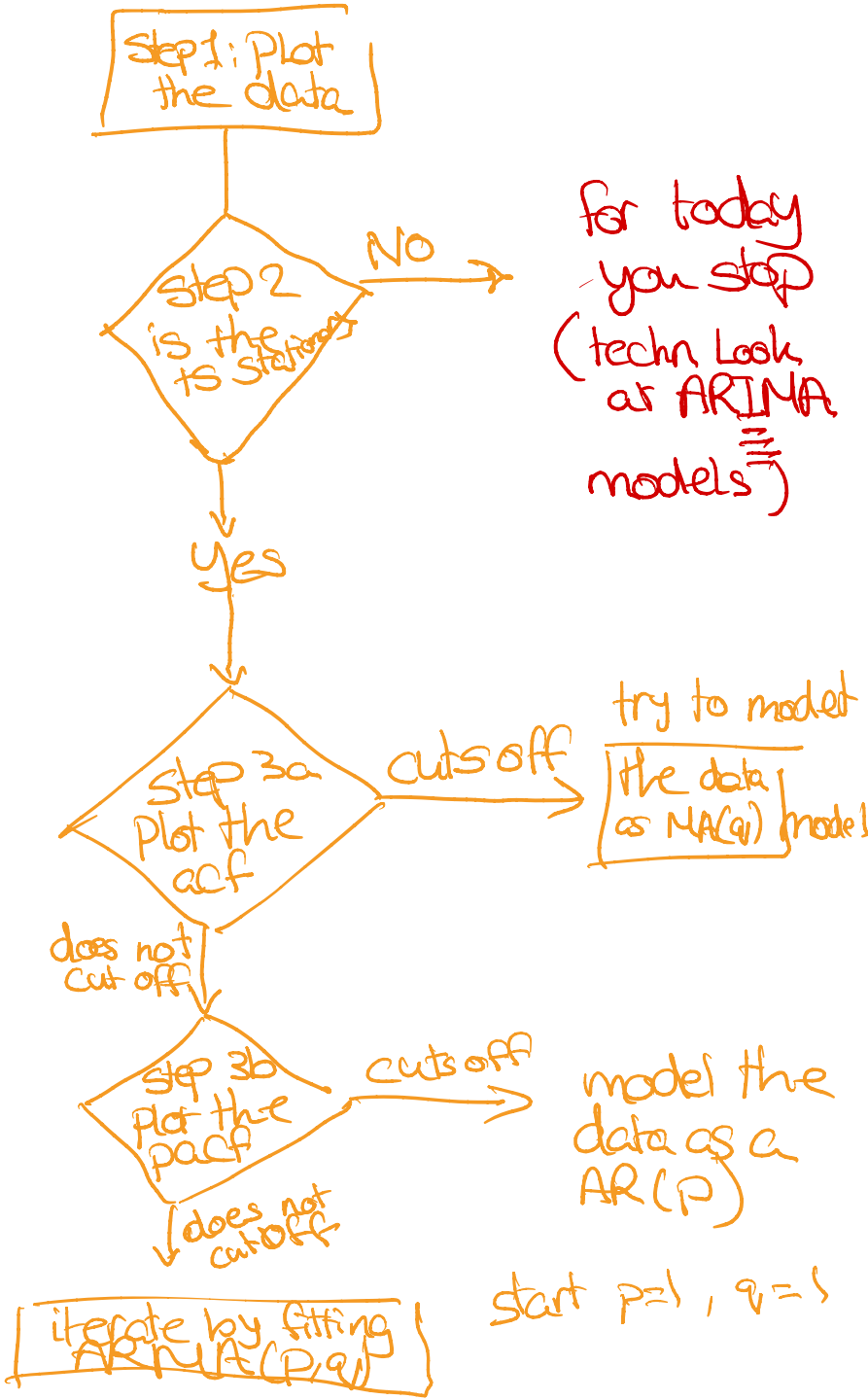
$\boxed{}$ → regression coeffient $* \ \varepsilon_{t-\square}$

→ weighted average of your errors for Lags 1 : q

population acf for MA(q)

$$\rho(k) = \begin{cases} \text{non-zero value for } k=1,\dots,q \\ 0 \text{ value for } k>q \end{cases}$$




Step 1: Plot the data
Step 2 is the ts stationary
No
for today you stop (techn Look at ARIMA models)
yes
Step 3a Plot the acf
cuts off
try to model the data as MA(q) model
does not cut off
Step 3b Plot the pacf
cuts off
model the data as a AR(p)
does not cutoff
iterate by fitting ARMA(p,q)
start p=1, q=1